Guangsheng Bao

PhD Candidate
Natural Language Processing
Westlake University and Zhejiang University

Advisor: Prof. Yue Zhang
baoguangsheng@westlake.edu.cn
Personal Website

Introduction

I am Guangsheng Bao, a Ph.D. candidate jointly enrolled at **Westlake University** and **Zhejiang University**, under the guidance of Prof. **Yue Zhang**. My primary research interests lie in the field of **trustworthy NLP**, encompassing topics such as model **generalization**, the creation of **explainable and controllable** NL models, ensuring the **safe application** of LLMs, and understanding the **reasoning capabilities** of LLMs. I have disseminated my research outcomes across a range of publications presented at esteemed conferences including ACL, ICLR, EMNLP, AAAI, and COLING. I have also actively participated in the academic community by serving as a reviewer for conferences like ACL, ICLR, EMNLP, ICML, NIPS, TACL, COLING, and others, and as an Area Chair for COLING 2022.

My research on **AI-generated text detection** was reported by *MIT Technology Review* and it has been productized (https://fastdetect.net/), with a visit volume of 20 millions in one year.

Prior to my pursuit of scientific research, I gained industry experience with positions at **Alibaba**, where I worked on customer service robot systems, and at **Microsoft** (China), where I contributed to the development of Cortana.

Publications

- **First/Co-first Author Papers** *2021-present*

  Guangsheng Bao, Yanbin Zhao, Juncai He, Yue Zhang. 2025. **Glimpse: Enabling White-Box Methods to Use Proprietary Models for Zero-Shot LLM-Generated Text Detection.** *In Proceedings of The Thirteenth International Conference on Learning Representations.* (**ICLR 2025, EI, CCF-A**)

  Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, Yue Zhang. 2024. **Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature.** *The Twelfth International Conference on Learning Representations.* (**ICLR 2024, EI, CCF-A**)

  Guangsheng Bao, Zhiyang Teng, and Yue Zhang. 2023. **Target-Side Augmentation for Document-Level Machine Translation.** *In Proceedings of the 61th Annual Meeting of the Association for Computational Linguistics.* (**ACL 2023, EI, CCF-A**)

  Guangsheng Bao, Yue Zhang, Zhiyang Teng, Boxing Chen and Weihua Luo. 2021. **G-Transformer for Document-Level Machine Translation.** *In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics.* (**ACL 2021 Oral, EI, CCF-A**)

  Guangsheng Bao and Yue Zhang. 2021. **Contextualized Rewriting for Text Summarization.** *In Proceedings of the AAAI Conference on Artificial Intelligence.* (**AAAI 2021, EI, CCF-A**)

  Guangsheng Bao*, Hongbo Zhang*, Linyi Yang, Cunxiang Wang, Yue Zhang. 2025. **How Likely Do LLMs with CoT Mimic Human Reasoning?** *In Proceedings of the 31st International Conference on Computational Linguistics.* (**COLING 2025, EI, CCF-B**)

  Hongbo Zhang*, Han Cui*, **Guangsheng Bao***, Linyi Yang, Jun Wang, Yue Zhang. 2025. **Direct Value Optimization: Improving Chain-of-Thought Reasoning in LLMs with Refined Values.** *In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing.* (**EMNLP 2025, EI, CCF-B**)

  Guangsheng Bao, Yue Zhang. 2023. **A General Contextualized Rewriting Framework for**

Text Summarization. *In IEEE/ACM Transactions on Audio, Speech and Language Processing.* (**TASLP 2023, SCI Q2, CCF-B**)

Guangsheng Bao, Zebin Ou, Yue Zhang. 2023. **GEMINI: Controlling The Sentence-Level Summary Style in Abstractive Text Summarization.** *In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing.* (**EMNLP 2023, EI, CCF-B**)

Guangsheng Bao*, Zhiyang Teng*, Hao Zhou, Jianhao Yan, Yue Zhang. 2023. **Non-Autoregressive Document-Level Machine Translation.** *In Findings of the 2023 Conference on Empirical Methods in Natural Language Processing.* (**EMNLP 2023 Findings, EI, CCF-B**)

Guangsheng Bao, Zhiyang Teng, and Yue Zhang. 2023. **Token-Level Fitting Issues of Seq2seq Models.** *In Proceedings of the 8th Workshop on Representation Learning for NLP.* (**RepL4NLP at ACL 2023**)

- **Under Review Papers** *2026-present*

  Guangsheng Bao*, Lihua Rong*, Yanbin Zhao, Xiao Yu, Qiji Zhou, Yue Zhang. 2026. **Triospect: A Three-Dimensional Framework for Robust Statistical AI-Generated Text Detection Against Diverse Attacks** (**TACL 2026 conditional accept**)

  Guangsheng Bao*, Hongbo Zhang, Han Cui, Yanbin Zhao, juncai He, Yue Zhang. 2026. **FAAST: Forward-Only Associative Learning via Closed-Form Fast Weights for Test-Time Adaptation** (**under review**)

  Zhizhang FU*, **Guangsheng Bao***, Hongbo Zhang, Chenkai Hu, Yue Zhang. 2026. **Correlation or Causation: Analyzing the Causal Structures of LLM and LRM Reasoning Process** (**under review**)

- **Coauthor Papers** *2020-present*

  Qiji Zhou*, Yifan Gong*, **Guangsheng Bao**, Hongjie Qiu, Jinqiang Li, Xiangrong Zhu, Huajian Zhang, Yue Zhang. 2025. **Reasoning is All You Need for Video Generalization: A Counterfactual Benchmark with Sub-question Evaluation.** *Findings of the Association for Computational Linguistics: ACL 2025.* (**ACL 2025 Findings**)

  Yixuan Weng*, Minjun Zhu*, **Guangsheng Bao**, Hongbo Zhang, Jindong Wang, Yue Zhang, Linyi Yang. 2025. **CycleResearcher: Improving Automated Research via Automated Review.** *In Proceedings of The Thirteenth International Conference on Learning Representations.* (**ICLR 2025**)

  Cunxiang Wang, Ruoxi Ning, Boqi Pan, Tonghui Wu, Qipeng Guo, Cheng Deng, **Guangsheng Bao**, Xiangkun Hu, Zheng Zhang, Qian Wang, Yue Zhang. 2025. **NovelQA: Benchmarking Question Answering on Documents Exceeding 200K Tokens.** *In Proceedings of The Thirteenth International Conference on Learning Representations.* (**ICLR 2025**)

  Linyi Yang, Shuibai Zhang, Zhuohao Yu, **Guangsheng Bao**, Yidong Wang, Jindong Wang, Ruochen Xu, Wei Ye, Xing Xie, Weizhu Chen, Yue Zhang. 2024. **Supervised Knowledge Makes Large Language Models Better In-context Learners** *The Twelfth International Conference on Learning Representations.* (**ICLR 2024**)

  Dandan Huang*, Leyang Cui*, Sen Yang*, **Guangsheng Bao**, Kun Wang, Jun Xie and Yue Zhang. 2020. **What Have We Achieved on Text Summarization?** *In Proceedings of the Conference on Empirical Methods in Natural Language Processing.* (**EMNLP 2020**)

Education

- **Westlake University, China** *Sep.2022 - present*
  Ph.D. candidate, natural language processing
- **University of Science and Technology of China** *Sep.2005 - Jan.2008*
  MEng, software engineering
- **HeFei University of Technology, China** *Sep.2000 – Jun.2004*
  BEng, computer science

Employment

- **AI Platform, Microsoft (US)** *Apr.2024 - Jun.2024*
  Research Internship Seattle
  – Research: automatic prompt optimization.
  – Submit an internal report that illustrates the 'lucky draw' behavior of current methods and demonstrate the empirical upper bound of the optimization.
- **Text Intelligence Lab, Westlake University** *Jun.2019 - Aug.2022*
  Research Assistant Hangzhou
  – Research: text summarization, neural machine translation, and coreference resolution.
  – Published two first-author long papers during the period, including AAAI 2021 and ACL 2021 (Oral).
- **Institute of Data Science and Technologies, Alibaba** *Mar.2015 - Jun.2017*
  Algorithm Expert Hangzhou
  – Research & development: online customer service robot system and hotline customer service robot system.
  – Investigated the latest Question Answering techniques and built them into the solution for customer service.
- **Information Platform Group, Microsoft (China)** *Spr.2013 – Feb.2015*
  Software Development Engineer Suzhou
  – Research & development: Natural Language Understanding (NLU) engine for Cortana.
  – Led several virtual teams working on training/testing/deployment tools and producing NLU models for languages such as Spanish, French, Italian, etc.
- **Exchange Server Group, Microsoft (China)** *Jan.2008 – Mar.2013*
  Software Development Engineer Beijing
  – Software design & development: Manageability for Exchange server.
  – Contributed to two complete release cycles of Exchange Server 2010 and Office 365.
- **Text-to-Speech Group, Microsoft (China)** *Jul.2006 – Sep.2007*
  Research Internship Beijing
  – Research: a mixture model for pitch contour prediction combining linear model and regression tree.
  – Improved the pitch prediction by about 13% and contributed the model to TTS engine.

Other Experience

- **Travel & Self-study in Edinburgh** *Jul.2017 - Jun.2018*
  Visitor Edinburgh
  – I accompanied my wife on a visiting scholarship at the University of Edinburgh during the period. I took the chance to explore my interests and studied some philosophy and cognitive science theories.
  – I explored cognitive science theories and tried to build a big picture of it, studying several topics such as sensation, perception, imagination, conception, and language.
  – I found my passion for these topics, so I decided to pursue a research opportunity on natural language processing.

Technical Skills and Interests

Languages: English (IELTS 7.0), Chinese (native speaker).
Developer Tools: Python, C++.
Soft Skills: Cross-team collaboration, Team leading.
Areas of Interest: AI, Robotics, Neural Science, Psychology.